Physician-Pharmacist Comanagement of Hypertension: A Randomized, Comparative Trial

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Objective. To compare the effectiveness of an evidence-based, systematic approach to hypertension care involving comanagement of patients by primary care physicians and clinical pharmacists versus usual care in reducing blood pressure in patients with uncontrolled hypertension.

Methods. Patients in a staff model medical group with uncontrolled hypertension were randomized to either a usual care (UC) or a physician-pharmacist comanagement (PPCM) group. All physicians in the study received both group and individual education and participated in the development of an evidence-based hypertension treatment algorithm. Physicians were then given the names of their patients whose medical records documented elevated blood pressures (defined as systolic ≥ 140 mm Hg and/or diastolic ≥ 90 mm Hg for patients aged < 65 yrs, and systolic ≥ 160 mm Hg and/or diastolic ≥ 90 mm Hg for those aged ≥ 65 yrs). Patients randomized to the UC group were managed by primary care physicians alone. Those randomized to the PPCM group were comanaged by their primary care physician and a clinical pharmacist, who provided patient education, made treatment recommendations, and provided follow-up. Blood pressure measurements, antihypertensive drugs, and visit costs/patient were obtained from medical records.

Results. One hundred ninety-seven patients with uncontrolled hypertension participated in the study. Both PPCM and UC groups experienced significant reductions in blood pressure (systolic -22 and -11 mm Hg, respectively, $p<0.01$; diastolic -7 and -8 mm Hg, respectively, $p<0.01$). The reduction in systolic blood pressure was greater in the PPCM group after adjusting for differences in baseline blood pressure between the groups ($p<0.01$). More patients achieved blood pressure control in the PPCM than in the UC group (60% vs 43%, $p=0.02$). Average provider visit costs/patient were higher in the UC than the PPCM group ($195 vs $160, $p=0.02$).

Conclusions. An evidence-based, systematic approach using physician-pharmacist comanagement for patients with uncontrolled hypertension resulted in improved blood pressure control and reduced average visit costs/patient.

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Nearly 25% of adults in the United States have hypertension; most individuals aged 65 years or older are among those affected.[1, 2] More than 50% of patients taking antihypertensive drugs have blood pressures that exceed recommendations, resulting in significant, preventable morbidity and mortality and contributing to the annual treatment costs of more than $10 billion.[3–6] Data from large randomized controlled trials show that even modest decreases in blood pressure (systolic

10–12 mm Hg, diastolic 5–6 mm Hg) can lead to significant reductions in the incidence of myocardial infarction and stroke over a 5-year period.[7, 8] Epidemiologic data suggest that many of the cardiovascular events avoided with improved hypertension management would occur in patients with only mildly elevated blood pressure.[9]

Numerous studies have documented quality-of-care issues in the management of hypertension and inadequate patient adherence to antihypertensive drug regimens.[10–21] One study of five Veterans Administration clinics found that less than 7% of hypertension-related visits resulted in an increase in antihypertensive drugs prescribed, even though 75% of patients had documented blood pressure measurements that exceeded national guidelines.[22] Efforts to improve management of hypertension by publishing guidelines over the past decade appear to have had little impact.[23, 24]

A systematic, comprehensive approach has been proposed as a method to improve quality of care for patients with chronic conditions in order to prevent common complications and disease progression.[25] This approach emphasizes using those interventions that have had the most significant impact on health outcomes. However, little data are available in the published literature that formally evaluate the effectiveness of this approach compared with that of other strategies for improving clinical and economic outcomes.[26] We performed a randomized, prospective trial in patients with uncontrolled hypertension to evaluate changes in blood pressure achieved by physician-pharmacist comanagement versus usual care.

Methods

Patient Identification and Randomization

Patients with capitated health insurance were recruited from the two main offices of a group medical practice of general internists and internal medicine subspecialists affiliated with a large community hospital. Baseline data were collected by chart review for all patients aged 18 years or older with capitated medical insurance and a diagnosis code for hypertension, according to claims and encounter data. Patients were excluded if advanced dementia, terminal illness, organ transplantation, or secondary hypertension was documented. Other exclusion criteria were absence of recorded blood pressure measurements, recent withdrawal from the medical group, insufficient or inadequate contact information (address and telephone number), recent change to noncapitated medical insurance, or missing medical records.

Elevated blood pressure was defined as systolic blood pressure 140 mm Hg or higher and/or diastolic 90 mm Hg or higher for patients younger than 65 years of age, and systolic 160 mm Hg or higher and/or diastolic 90 mm Hg or higher for patients aged 65 years or older.[23] A diagnosis of hypertension was confirmed by chart review. Patients were considered to have uncontrolled hypertension if the last two blood pressure measurements documented within the previous 3 months were elevated. If two blood pressure measurements were not documented within this period, patients were classified as uncontrolled if the last recorded blood pressure measurement was elevated. Patients with uncontrolled hypertension were then randomized to either a usual care (UC) or a physician-pharmacist comanagement (PPCM) group.

Intervention

An evidence-based treatment algorithm for management of hypertension was developed by a multidisciplinary team of physicians, clinical pharmacists, and nurses using the (then current) fifth report of the Joint National Committee on Detection, Evaluation, and Treatment of High Blood Pressure (JNC V).[23] The guideline was presented to and modified by participating study physicians after a discussion of published literature regarding hypertension epidemiology and associated morbidity and mortality. The final guideline was used as the basis for group education in subsequent physician meetings and in individual physician education sessions conducted by clinical pharmacists and the principal investigator (JEB).

Physicians were given the names of their patients whose medical records documented

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elevated blood pressures according to the hypertension algorithm and study criteria. After receiving their physician's permission, patients were contacted by phone and asked to participate in the study.

No intervention was made for patients randomized to the UC group. Those randomized to the PPCM group were asked to attend a hypertension clinic run by clinical pharmacists. In this clinic, pharmacists determined blood pressure with an automated blood pressure cuff calibrated to a mercury sphygmomanometer. They also collected patient assessments to determine adherence to antihypertensive drugs, potential drug side effects, and relevant patient habits, such as tobacco use, diet, and exercise, in accordance with JNC V guidelines. In addition, during the clinic, the pharmacists reviewed common drug side effects and provided individualized education regarding dietary and lifestyle modifications.

According to protocol, pharmacists then called each patient's physician, or a covering physician, with their findings and made treatment recommendations in accordance with the evidence-based treatment algorithm. Suggesting changes in antihypertensive drugs for the sole purpose of reducing treatment costs (i.e., switching to a less costly alternative between two equivalent agents) was not permitted, as the primary study end point was blood pressure control. Physicians made all final treatment decisions. Follow-up visits were scheduled every 2–4 weeks at the discretion of the pharmacist.

In this study, the goal of hypertension treatment was achievement of reduced blood pressures, based on common practice when the study began: systolic pressure below 140 mm Hg and diastolic pressure below 90 mm Hg for patients younger than 65 years, and systolic below 160 mm Hg and diastolic below 90 mm Hg for patients aged 65 years or older. Patients were seen as often as deemed necessary by the clinical pharmacist until blood pressure control was achieved; this was defined as two consecutive visits with blood pressure values at goal levels, at which time patients were discharged from the clinic. Patients' access to their primary care physicians was not restricted in any way during the study.

Outcomes Measures and Statistical Analysis

The primary study outcome was the difference between the PPMC and UC groups in changes in blood pressure over 12 months. Secondary outcome measures were the differences between the two study groups in the proportion of patients achieving goal blood pressure, costs of antihypertensive drugs, and total provider visit costs.

All patients were analyzed in the group to which they were randomized (intent to treat) unless they met one or more of the following exclusion criteria: their physician refused permission for patient contact, the patient refused to participate in the study, one or no prospective blood pressure measurements were documented in the patient's medical chart during the study period, the first (prospective) blood pressure measurement after telephone contact was not elevated according to study criteria, and new information obtained during telephone contact made the patient ineligible (e.g., recently changed to noncapitated insurance).

Changes in blood pressure were determined by comparing each patient's first (prospective) measurement recorded after the date of telephone contact with the last measurement recorded over the next 12 months. A similar procedure was used to calculate changes in blood pressure at 3, 6, and 9 months from baseline. Hypertension control rates were evaluated using the χ^2 test. Differences in continuous variables were determined using the Student *t* test. Analysis of covariance was used to adjust for differences in baseline blood pressure measurements in determining the differences in blood pressure changes between the UC and PPCM groups.

The perspective of the economic analysis was that of a capitated medical group that was also at risk for prescription drug costs. Drug costs were determined using average wholesale prices, with no discount, for the antihypertensive drugs prescribed. Final drug costs were defined as the sum of costs for all antihypertensive drugs at the time of the last provider visit (either physician or pharmacist) within the study period. Provider visit costs were estimated at $20 for a 30-minute pharmacist appointment and $35 for a 15-minute physician appointment. These costs were based on salary levels of physician, pharmacist, and office personnel at the time of the study.

Results

A total of 1272 patients with hypertension who met the initial inclusion criteria were identified from 1996–1998 and were randomized to either the UC group (637 patients) or the PPCM group

(635 patients). Of the patients randomized, 538 in the UC group and 537 in the PPCM group met at least one of the exclusion criteria after prospective data collection. The most common reasons for excluding patients after randomization were withdrawal before the study began and having noncapitated insurance (160 and 173 patients in the UC and PPCM groups, respectively). In addition, 114 UC and 31 PPCM patients were excluded due to lack of contact information, and 14 UC and 13 PPCM patients were excluded due to comorbidities (advanced dementia, terminal illness, or organ transplantation). The remaining excluded patients had blood pressures below study entry criteria, either on confirmatory chart review performed immediately before patient contact (15 UC and 31 PPCM patients) or on first measurement at study entry (70 UC and 45 PPCM patients).

A total of 99 UC and 98 PPCM patients were eligible for analysis (Figure 1). Baseline characteristics of the study patients are provided in Table 1. No statistically significant differences between the two groups were observed for age, sex, baseline diastolic blood pressure, or comorbid conditions (e.g., coronary artery disease, chronic heart failure, and diabetes mellitus). However, the PPCM group had more African-American patients (40 vs 27, p=0.05) and higher systolic blood pressure at baseline (162 vs 156 mm Hg, p=0.02) than the UC group. No statistically significant differences between the two groups were observed for demographics, comorbid conditions, or baseline blood pressures for patients who were eligible but refused to participate in the study (92 in UC group, 206 in PPCM group).

Of the 168 PPCM patients who agreed to participate, 143 (85.1%) attended the pharmacist-run clinic; 45 (31.5%) of these had blood pressures below study entry criteria at their initial visit versus 70 of 169 (41.5%) UC patients who agreed to participate. Four clinical pharmacists participated in the PPCM intervention. Of the 40 physicians asked to participate in the study, one refused. Median time

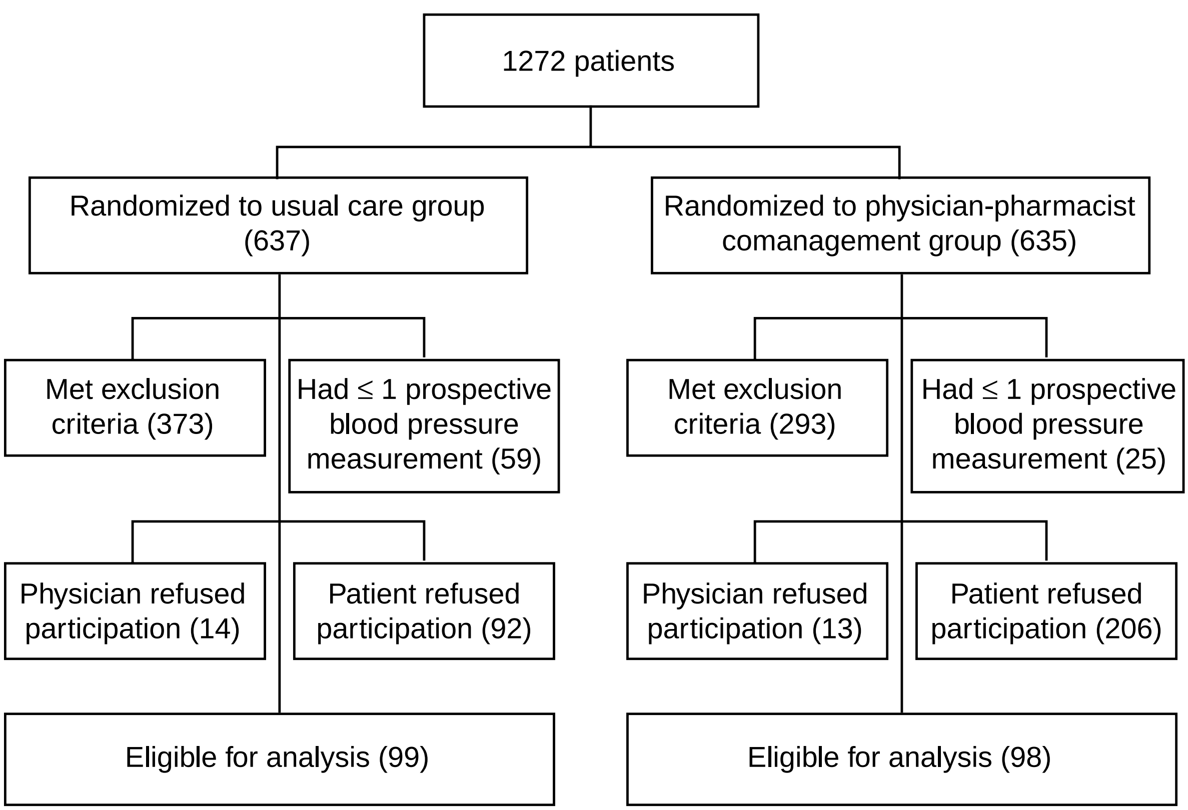

Figure 1. Patient randomization to the usual care and physician-pharmacist comanagement study groups. Prospective data collection identified additional patients who met the exclusion criteria. Study inclusion further required permission of the patient's primary care physician to contact the patient, and agreement of the patient to participate in the study. Two or more prospective blood pressure measurements were required for analysis of individual patient data.

Table 1. Baseline Characteristics of Patients with Hypertension

	UC Group (n=99)	PPCM Group (n=98)
M/F	41/58	36/62
Age		
Mean (yrs)	61.5	62.5
≥ 65 yrs (no. of pts)	35	31
African-American (no. of pts)[a]	27	40
Mean blood pressure (mm Hg)		
Systolic[a]	156	162
Diastolic	90	92
Mean antihypertensive drug costs/mo ($)	24.46	28.59
Medical conditions (no. of pts)		
Angina	1	1
Dyslipidemia	67	55
Asthma or COPD	5	6
Coronary artery disease	2	3
Previous stroke	5	5
Chronic heart failure	5	1
Previous myocardial infarction	2	2
Diabetes mellitus	13	13
Renal impairment	2	1

UC = usual care; PPCM = physician-pharmacist comanagement; COPD = chronic obstructive pulmonary disease.
[a]Statistically significant (p<0.05).

elapsed was 76 days between the first and last pharmacist visits for PPCM patients who were seen in the clinic.

Clinical Outcomes

At 12 months, reductions in systolic blood pressure from baseline for the PPCM and UC groups were 22 mm Hg (p<0.01) and 11 mm Hg (p<0.01), respectively. The greater reduction of 10 mm Hg in systolic blood pressure observed in PPCM versus UC patients was statistically significant (p<0.01). This difference persisted after adjustment for baseline blood pressure using analysis of covariance (Figure 2). Reductions in diastolic blood pressure from baseline for the PPCM and UC groups were 7 mm Hg (p<0.01) and 8 mm Hg (p<0.01), respectively. The between-group difference in diastolic blood pressure was not statistically significant (p=0.53).

Overall, blood pressure goals were achieved in 60% and 43% of PPCM and UC patients, respectively (p=0.02). The percentage of patients receiving at least one first-line antihypertensive agent according to the evidence-based algorithm increased significantly from baseline in both the PPCM group (from 68% to 80%, p=0.02) and the UC group (from 60% to 70%, p=0.02).

Economic Outcomes

The average provider visit costs/patient were lower for PPCM than UC patients ($160 vs $195, p=0.04). This difference resulted from a lower

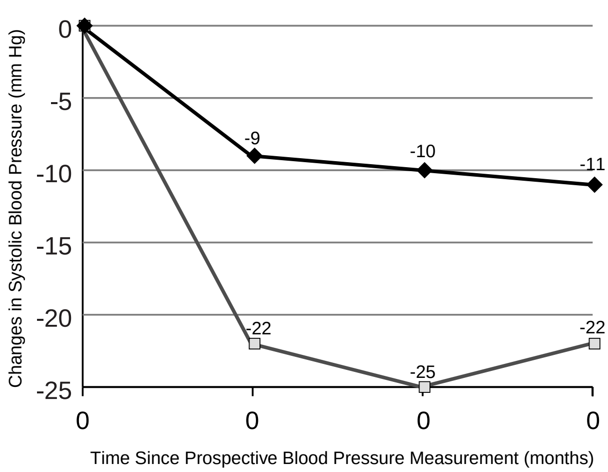

Figure 2. Changes in systolic blood pressure over time in the usual care (UC) group (◆, 99 patients) versus the physician-pharmacist comanagement (PPCM) group (❏, 98 patients), after adjusting for baseline blood pressure differences using analysis of covariance. The mean reduction in systolic blood pressure was greater in PPCM than UC patients at 6, 9, and 12 months (p<0.01 for all comparisons).

average number of visits to the primary care physician during the study by PPCM than UC patients (3.4 vs 6.6, $p<0.01$). However, there was a strong trend toward more total provider (physician and pharmacist) visits in the PPCM versus UC patients (8.0 vs 6.6, $p=0.06$). Although no statistically significant differences were noted in average monthly drug costs between the two patient groups at the end of the study period, a trend toward a greater increase in drug costs from baseline was observed in the PPCM versus the UC group ($11.31 vs $4.25, $p=0.12$).

Discussion

Our findings show that physician-pharmacist comanagement of hypertension, using an evidence-based treatment algorithm, led to higher rates of blood pressure control and greater systolic blood pressure reductions in patients with uncontrolled hypertension than usual care by primary care physicians alone. Annual provider visit costs and average number of physician visits were lower for the PPCM group, although the number of annual total provider (physician and pharmacist) visits did not differ significantly between the groups.

The changes in systolic blood pressure from baseline in both groups were statistically and clinically significant, generally meeting or exceeding the magnitude of reduction shown in systematic reviews of randomized controlled trials to reduce morbidity and mortality significantly over 5.4 years.[7, 8] The significantly greater reduction in systolic blood pressure in the PPCM versus UC patients is likely clinically relevant as well, even though changes in diastolic blood pressure in the two groups were similar. According to the Framingham cohort and other population databases, systolic blood pressures are more highly correlated with adverse events than diastolic blood pressures.[6, 19, 21] However, despite the greater reduction in systolic blood pressure observed in the PPCM group, the significant proportion (40%) of patients who failed to achieve blood pressure control clearly illustrates the limitations of this intervention.

The reason for the difference between the two groups in number of total provider visits is unclear but may reflect more effective follow-up of PPCM patients. Achieving improved follow-up of patients with hypertension by involving clinical pharmacists could have important economic implications. The annual rate of ambulatory patient visits in the United States is higher for hypertension than for any other chronic condition.[2] Lower annual provider visit costs and number of physician visits in the PPCM group could have significant implications for provider access and hypertension treatment costs in similar settings.

The trend toward greater increases in drug costs in the PPCM group, although not statistically significant, could offset other savings. However, because this study focused on improving clinical outcomes for patients with hypertension, decisions to change drugs solely based on economic considerations were not permitted for the PPCM group. The absence of a similar restriction for the UC group could have led to lower antihypertensive drug costs.

Patients enrolled in this study had elevated blood pressure. Considering the deleterious effects of uncontrolled hypertension, the study design incorporated interventions intended to ensure appropriate care for all study patients. Systematic reviews of interventions directed at changing physician behavior indicate that both group and individual physician education along with providing physicians with pertinent clinical information about individual patients, as in this study, are modestly effective.[27, 28]

Our findings suggest, however, that the addition of clinical pharmacists to aid in patient management, guided by an evidence-based algorithm, resulted in an incrementally greater improvement in blood pressure control. The results are consistent with those from other studies of integrated programs for the care of patients with hypertension.[29–32] Collectively, these studies support the model for an evidence-based, systematic approach,[25] which suggests that multiple simultaneous interventions at the patient and provider level should be more efficacious than those directed at either patients or providers alone.

The cost-effective management of hypertension by clinical pharmacists has been demonstrated.[33] One important aspect of our study is that hypertension management was improved in a typical clinical setting: the offices of a staff-model medical group. The relevance of interventions that are effective in common practice venues is supported by a recent report based on an evaluation of medical records from a large health system.[34] The report found that 60% of patients with hypertension had blood pressures that exceeded recommended levels. Other studies suggest that different models of physician comanagement may be effective, ranging from the use of pharmacists to aid in screening for uncontrolled hypertension[35, 36] to active collaboration in treatment.[32] Although many barriers hinder the implementation of such programs, direct clinical interventions by pharmacists appear to be welcomed by patients, and may even be preferred over typical community pharmacist interactions.[37]

One limitation of our study was the exclusion of patients after randomization, particularly in the PPCM group. We found no difference in baseline characteristics between patients who agreed to participate and those who refused. Potential selection biases, such as patient motivation or commitment to achieving blood pressure control resulting from different postrandomization exclusion rates between the UC and PPCM groups cannot be discounted. This problem was anticipated when the study was designed but was considered unavoidable since allowing patients to know the details of the intervention to which they were randomized before requesting their participation was considered, by participating physicians, the most compatible with real-world practice. In addition, the established goal for systolic blood pressure (160 mm Hg for patients aged 65 yrs or older), although considered reasonable at the beginning of the study, is higher than current recommendations.[6]

Another limitation was the difference in systolic blood pressures between the two groups at baseline. Consequently, comparing changes in systolic blood pressure required an analysis of covariance. However, a greater proportion of PPCM than UC patients achieved blood pressure control. Lack of a control group confounds the interpretation of within-group changes in blood pressure from baseline, although the magnitude of the changes observed are similar to those seen in a controlled study of a pharmacist-run clinic for patients with hypertension.[32] The greater reduction in systolic blood pressure observed in PPCM patients could be explained, at least in part, by the higher frequency of provider visits in this group, although this difference did not reach statistical significance.

In our study, the PPCM group achieved significant blood pressure reductions; however, we did not measure clinical outcomes associated with uncontrolled hypertension, such as increased frequency of stroke and myocardial infarction. Improvements in these outcomes would require persistence of the blood pressure reductions over a prolonged period.

Drug costs in our study were projected based on the average wholesale prices of the prescribed regimen. Actual drug costs, which reflect patient compliance rates and acquisition costs that can vary by payor, were not determined. Similarly, visit costs/patient were projected from salaries of physician, pharmacist, and office personnel. In a capitated health insurance environment, these salaries represent fixed costs. Actual savings in visit costs through the use of clinical pharmacists to comanage patients with hypertension would occur only if the resulting reduction in physician workload allowed individual physicians to care for more patients.

Conclusion

Compared with usual care, physician-pharmacist comanagement of patients with uncontrolled hypertension resulted in greater reductions in systolic blood pressure, a larger proportion of patients achieving blood pressure control, and reduced provider visit costs, with no increase in antihypertensive drug costs. Additional research of physician-pharmacist comanagement of hypertension in other patient populations and various clinical settings are required to better assess the ability to generalize these findings.

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